Math 10C — Unit 8: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

1. Consider the system of equations $x + 3y = 1,\ \ 2x - y = 9$.

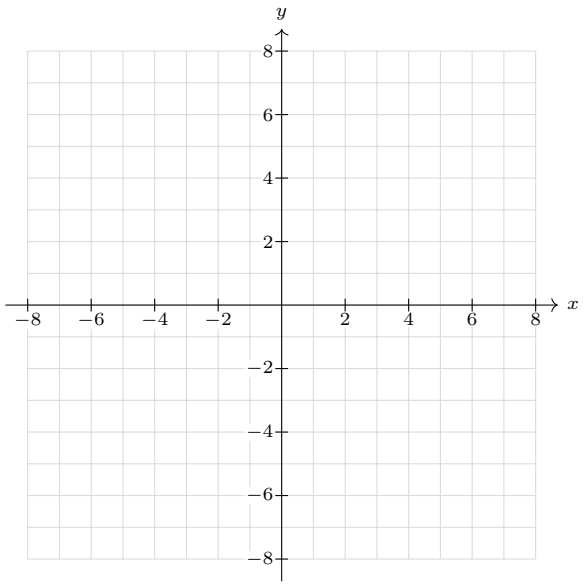
(a) Write each equation in slope–intercept form.

(b) Complete the table of values for each equation.

$x + 3y = 1$		$2x - y = 9$	
x	y	x	y
-5		2	
-2		3	
1		4	
4		5	
7		6	

(c) Solution: _____

(c) Draw both lines on the grid and state the solution to the system.



(d) Verify the solution algebraically.

2. Both equations in each system are in slope–intercept form. Solve each system by setting the two expressions for y equal (Section 3), then verify the solution by substituting into both original equations.

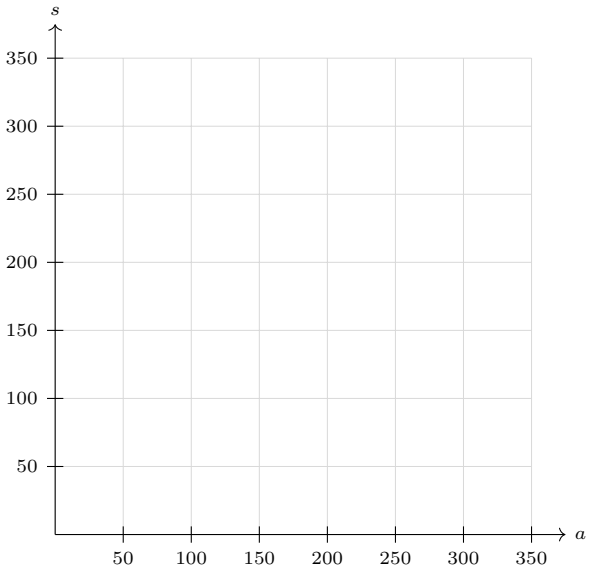
(a) $y = 2x - 5,\ \ y = -x + 4$ (b) $y = -2x,\ \ y = -\frac{1}{4}x - 7$ (c) $y = x + 3,\ \ y = \frac{2}{3}x - 1$

3. All 350 tickets for a school play were sold. Adult tickets cost \$8 each, and student tickets cost \$5 each. Total receipts were \$2350. The information from the number of tickets can be represented by $a + s = 350$.

(a) State an equation which can be formed from the costs of the tickets.

(b) Graph the system to determine the number of tickets sold of each type.

Adult tickets: _____
Student tickets: _____

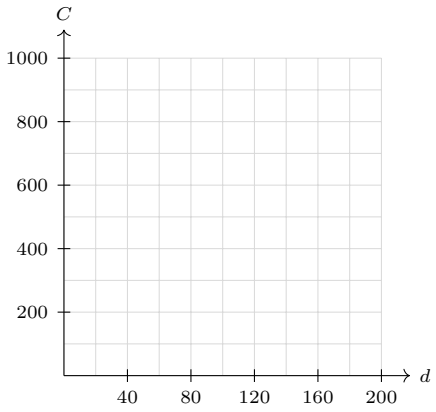


(c) Verify your solution algebraically.

4. A school is choosing a bus company for a field trip. Company A charges \$350 plus \$2.50 per kilometre. Company B charges \$5.00 per kilometre. Let C be the cost in dollars for a trip of d kilometres.

(a) Write a cost equation for each company.

(b) Graph both equations on the grid, without technology.

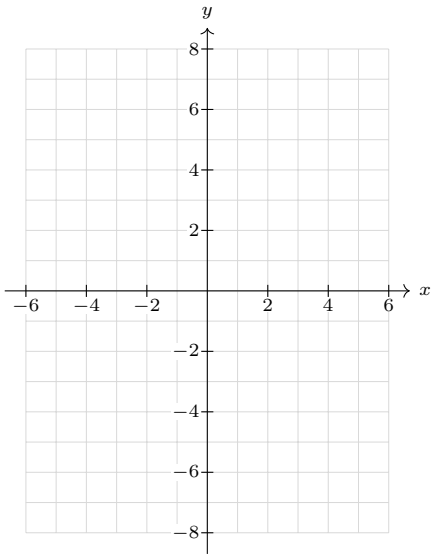


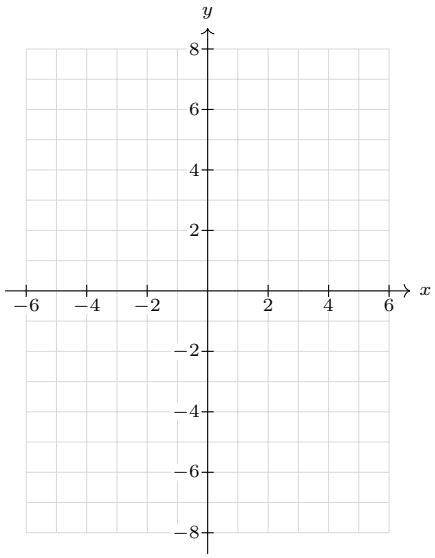
(c) State the point where the two lines cross, and explain what it means for the school.

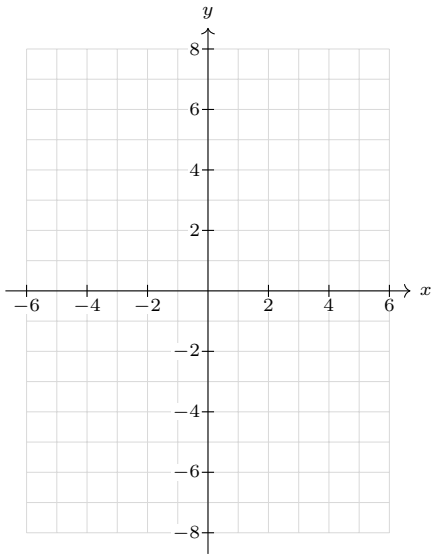
(d) For which trip lengths is Company B cheaper?

5. Graph each system and determine whether it has no solution, one solution, or infinitely many solutions.

(a) $y = 3x - 1,\ \ y = -2x + 9$ (b) $y = -x + 4,\ \ y = -x - 2$ (c) $4x - 2y = 8,\ \ y = 2x - 4$







6. Rearrange each equation into the form $y = mx + b$ and state whether the system has no solution, one solution, or infinitely many solutions.

(a) $4x - y = 7,\ \ y = 4x + 7$ (b) $9x - y = 2,\ \ x - 9y = 2$ (c) $4y + x - 8 = 0,\ \ y = -\frac{1}{4}x + 2$

7. Write an equation which forms a system with the equation $5x - y = 3$ so that the system has:

(a) no solution (b) one solution (c) infinitely many solutions

8. *(Multiple Choice)* The system $\begin{cases} 6x - 4y = 10 \\ 9x - 6y = 12 \end{cases}$ has

A. no solution
B. one solution, $(3, 2)$
C. one solution, $(1, -1)$
D. infinitely many solutions

9. *(Numerical Response)* The value of k, where k is a natural number, for which the system of equations $6x + ky = -12$ and $-9x - 3y = 18$ has an infinite number of solutions, is _____.

Record your answer left to right. ☐☐☐☐

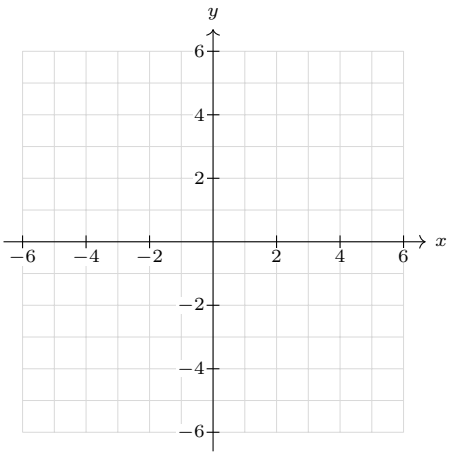
10. *(Numerical Response)* The value of a, where a is a natural number, for which the system of equations $ax + 6y = 12$ and $4x + 3y = 9$ has no solution, is _____.

Record your answer left to right. ☐☐☐☐

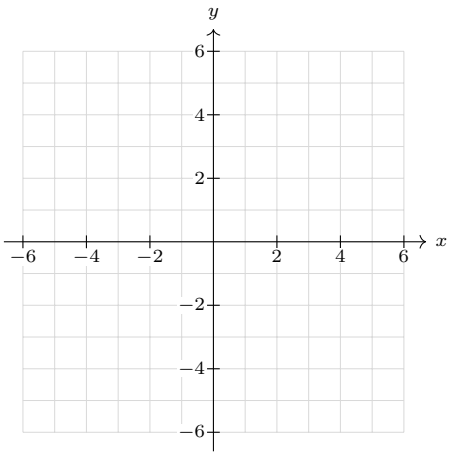
Lesson 2 Assignment *Show your work.*

1. Solve each system using the method of substitution and verify that the solution satisfies both equations. Then check each solution by graphing both lines on the grid below it.

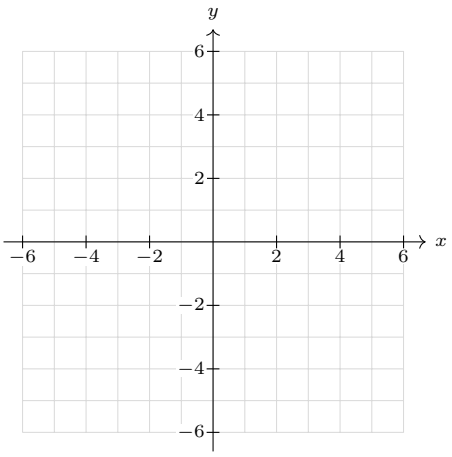
(a) $3x - 2y = 6,\ \ \frac{1}{2}x - y = -1$



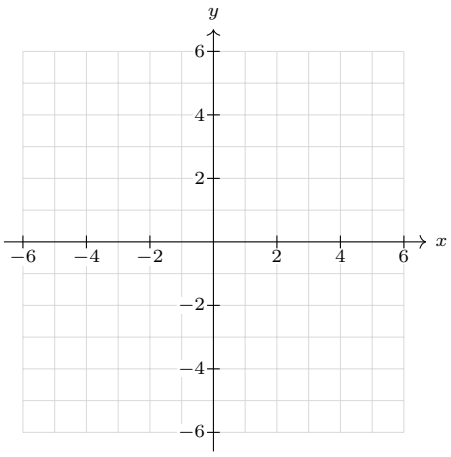
(b) $3(x + 1) + y = -1,\ \ 5x - 2(y - 4) = -17$



(c) $3(x - 2) + 2(y + 4) = 2,\ \ 5x - 3(y + 2) = 13$



(d) $3(2x + 1) - (y - 5) = -2,\ \ 4(1 - 3x) - 3(5 - 2y) = 25$



2. Solve each of the following systems by substitution. Check each solution.

(a) $y = x + 5,\ \ 2x + 3y = 5$

(b) $x - 3y = 11,\ \ x + 2y - 6 = 0$

(c) $3p + q = 0,\ \ 5p + 2q = -1$

(d) $4u - 2v + 8 = 0,\ \ 2u = 2v - 3$

(e) $4x + 3y = 7,\ \ x - y = 0$

(f) $0.5x + 0.4y = 1.7,\ \ 2x - 3y = 16$

3. Solve each system by substitution. Give the answers as exact values (fractions, not decimals).

(a) $4x - y = 6,\ \ y = -2x + 3$

(b) $0.2x + 0.5y = 1.9,\ \ 3x - y = -0.4$

4. The straight line $px + qy + 10 = 0$ passes through the points $(1, 4)$ and $(-2, 2)$.

(a) Substitute the x- and y-coordinates of the two points into the equation of the line to form two equations in p and q.

(b) Solve this system of equations by substitution to determine the values of p and q, and write the equation of the line.

(c) Verify the equation in (b) using the slope formula and the slope–point form of the equation of a line.

5. Solve the following systems by substitution. Explain the results.

(a) $y = 2x - 5,\quad 4x - 2y = 10$

(b) $x = 2y + 3,\quad 3x - 6y = 4$

6. A notebook costs \$3 less than two pens. Five pens cost the same as three notebooks. Set up and solve a system of equations to determine the cost of a notebook.

7. *(Multiple Choice)* If $x + 3y = 17$ and $x - 3y = -1$, then $x + y$ is equal to
A. 11 B. 14 C. 5 D. -3

8. *(Multiple Choice)* When solving a system of equations, one of which is $\frac{x}{3} - \frac{y}{4} = 1$, a substitution which can be made is
A. $x = \frac{1}{4}(4y + 12)$ B. $y = \frac{1}{4}(3x - 12)$
C. $x = \frac{1}{4}(3y + 12)$ D. $y = \frac{1}{3}(4x + 12)$

9. *(Numerical Response)* If $2m - 5n + 8 = 3m - 4n - 1 = 0$, then the value of $m - n$, to the nearest tenth, is _____.
Record your answer left to right. ☐☐☐☐

10. *(Numerical Response)* Ben pays \$23 for 3 movie tickets and 2 bags of popcorn. Mia pays \$15 for 2 movie tickets and 1 bag of popcorn. The cost of one movie ticket, in dollars, is _____.
Record your answer left to right. ☐☐☐☐

11. *(Extension)* Solve each of the following systems using substitution.
(a) $y = 2x, \quad 3x - y = 7, \quad x + 2z = 9$
(b) $x = 2y - 2, \quad y + 3z - 2 = 0, \quad x - 4z = -8$

Lesson 3 Assignment *Show your work.*

1. Solve each system by **adding** the two equations, then verify the solution satisfies both equations.

(a) $3x + 4y = 26$
$5x - 4y = 22$

(b) $x + 5y = -8$
$3x - 5y = 36$

(c) $x + 2y = -6$
$-x + 4y = -18$

(d) $5a - 2b = 19$
$-5a - 3b = -9$

2. Solve each system by **subtracting** the two equations, then verify the solution satisfies both equations.

(a) $6x + y = 29$
$2x + y = 13$

(b) $4m + 5n = 32$
$4m - 3n = 0$

(c) $-x - 6y = -13$
$-x + 2y = 11$

(d) $3a - 2b = -24$
$-3a - 2b = 0$

3. Solve each system by elimination. In each, **one** equation needs to be scaled. Check each solution.

(a) $a + 3b = 11$
$2a - b = 8$

(b) $5x - 2y = 13$
$3x - 4y = 5$

(c) $4x - 3y = 0.5$
$3x + y = 2$

4. Solve each system by elimination. In each, **both** equations need to be scaled. Check each solution. Give exact values (fractions) where necessary.

(a) $2x + 3y = 12$
$3x - 4y = 1$

(b) $4x + 3y = -1$
$5x + 2y = 4$

(c) $3x + 2y = 7$
$5x - 3y = 1$

5. Solve each system by elimination. Check each solution.

(a) $3x + 2y = 8.5$
$5x - 4y = 10.5$

(b) $7x = 4y$
$3x - 2y + 2 = 0$

(c) $4e + 3f - 11 = 0$
$3e + 2f - 9 = 0$

(d) $4x + 3y - 2 = 0$
$8x = 5y + 26$

6. Solve the system $x - 2y - 3 = 0, \quad 3x + 4y = 19$ by each method below.
(a) elimination (b) substitution (c) Which do you prefer? Why?

7. Solve the system $9x + 2y - 34 = 0, \quad 9x - 4y - 94 = 0$ by each method below.
(a) elimination (b) substitution (c) Which do you prefer? Why?

8. Solve each system by elimination. Explain the result in each case.
(a) $-3x + 9y - 6 = 0, \quad 4x - 12y + 8 = 0$ (b) $2x - 5y + 3 = 0, \quad -4x + 10y + 9 = 0$

9. Solve each system by elimination.
(a) $2x - \frac{y}{3} = 9$
$\frac{x}{2} + \frac{y}{4} = \frac{21}{4}$
(b) $\frac{m}{3} - \frac{n-2}{4} = 2$
$\frac{m}{2} - \frac{n}{3} = \frac{5}{2}$
(c) $\frac{1}{3}(3x - 2y) + \frac{1}{4}x = 8$
$\frac{1}{4}x - \frac{1}{2}y = \frac{1}{2}$

10. *(Multiple Choice)* When y is eliminated from the equations $3x + 2y = 11$ and $5x - 4y = 9$, we obtain
A. $8x = 20$ B. $11x = 31$ C. $x = -2$ D. $2x = 9$

11. *(Multiple Choice)* The solution to the system $x - y = 0, \ \frac{1}{3}x + \frac{1}{4}y = 7$ is
A. $x = 12, \ y = 12$ B. $x = -12, \ y = -12$
C. $x = 0, \ y = 0$ D. $x = 7, \ y = 7$

12. *(Multiple Choice)* The ordered pair (x, y) which satisfies the system of equations $x - 3y = 9, \ x + 2y = 4$ is
A. $(6, -1)$ B. $(-1, 6)$ C. $(-6, -1)$ D. $(6, 1)$

13. *(Numerical Response)* If $\frac{1}{4}x + 3 = \frac{1}{2}y$ and $\frac{1}{3}x - \frac{1}{4}y = -1$, then the value of $x + 2y$ is _____.

Record your answer left to right. | | | | |

14. *(Numerical Response)* If $3x + 2y = 44$ and $x - 2y = -4$, then the value of $x + 3y$ is _____.

Record your answer left to right. | | | | |

Lesson 4 Assignment *Show your work.*

For questions 1–9: introduce variables (with units), form a system of two equations, solve the system, and answer the question in a sentence.

1. A rectangle is to be drawn with perimeter 70 cm. If the length is to be 11 cm more than the width, determine the area of the rectangle.

2. The sum of two numbers is 7. Twice the larger number is 39 more than three times the smaller number. Find the numbers.

3. Six notebooks and three folders cost \$28.50. Four similar notebooks and seven similar folders cost \$30.50. **Describe** a method to determine how much you would expect to pay for ten notebooks and five folders, and then calculate the cost.

4. The perimeter of a rectangle is 50 cm. If the length were doubled and the width halved, the perimeter would be increased by 20 cm. Find the dimensions of the original rectangle.

5. At a theatre, a main-floor seat costs twice as much as a balcony seat. If five main-floor seats cost \$54 more than eight balcony seats, find the cost of a main-floor seat.

6. Jamal has been saving nickels and quarters. He has 86 coins in total, worth \$15.50. How many coins of each type does he have?

7. One year, a woman's savings (income − expenditure) were \$4200. The next year her income increased by 10% and her expenditure decreased by 15%, allowing her to save \$13 570. Determine her income in the first year.

8. Priya invested $\frac{5}{8}$ of her \$64 000 inheritance in two mutual funds. By the end of the year, the *Growth Fund* had gained 9% and the *Bond Fund* had lost 2%. If the combined value of her investments increased by \$1950, determine the amount invested in each fund.

9. Ken invested \$9000, part at 8% interest and part at 5% interest. The interest earned from the 5% portion was half of the interest earned from the 8% portion. How much was invested at each rate?

10. *(Multiple Choice)* A school sold 300 tickets to a concert. Student tickets cost \$6 and adult tickets cost \$10, and the ticket sales totalled \$2240. If s is the number of student tickets and a the number of adult tickets, which system models the situation?

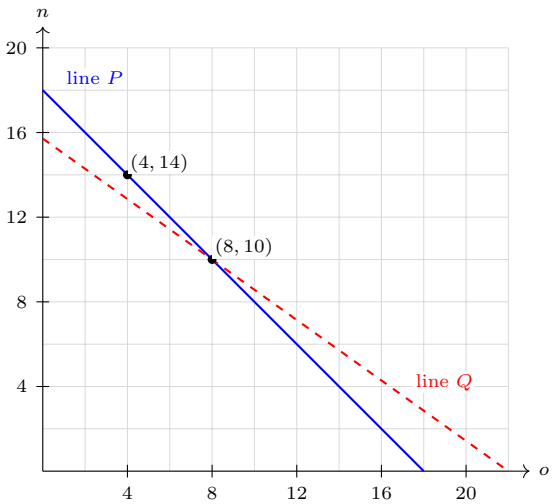
A. $s + a = 300,\ \ 6s + 10a = 2240$

B. $s + a = 2240,\ \ 6s + 10a = 300$

C. $s + a = 300,\ \ 10s + 6a = 2240$

D. $s + a = 16,\ \ 6s + 10a = 2240$

11. The graph shows the system from Example 5: an older machine packs 25 boxes per hour and a newer machine 35 boxes per hour; o and n are the hours each one ran.



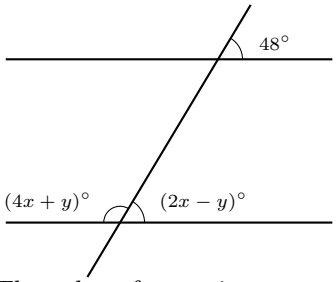
(a) Line P is $o + n = 18$ and line Q is $25o + 35n = 550$. Which line shows the total *time* and which the total *output*?

(b) What does the point $(8, 10)$ mean in this context?

(c) The point $(4, 14)$ lies on line P only. What does it mean, and why is it not the solution?

12. A theatre clerk reports that 40 tickets were sold — adult tickets at \$12 and child tickets at \$7 — for a total of \$393. Set up and solve a system. Does the answer make sense? What can you conclude about the clerk's report?

13. *(Numerical Response)* In the diagram, two parallel lines are cut by a transversal. The angle marked $48°$ and the angle marked $(2x - y)°$ are corresponding angles (so they are equal), and the angles marked $(2x - y)°$ and $(4x + y)°$ lie on a straight line (so they add to $180°$).



The value of $x + y$ is _____.

Record your answer left to right. | | | | |

14. *(Numerical Response)* A two-digit number has digits whose sum is 13. If the digits are reversed, the new number is 45 more than the original number. The original number is _____.

Record your answer left to right. | | | | |

15. *(Extension)* The heights, in metres, of the vertical support rods of a footbridge canopy, measured as you move out from the centre, form the sequence 0.7, 1.3, 2.3, 3.7, ...

(a) Without a calculator, determine the next two terms of the sequence.

(b) The height, h metres, of the nth rod is given by $h = a + bn^2$. Using two terms of the sequence, form a system of equations and determine a and b.

(c) Use your formula from part (b) to verify your answers in part (a).

Lesson 5 Assignment *Show your work.*

For every question: introduce a variable for each unknown *with its unit*, form a system of two equations, solve it, check the solution against both original conditions, and answer in a sentence. Questions 1–4 and 8 are mixtures; Questions 5–7 and 9 are distance-speed-time.

1. Granola bars costing \$5.50 per kg are mixed with granola bars costing \$3.50 per kg to produce 96 kg of mixture worth \$438. How many kg of each type of granola bar were used?

2. 400 g of Type X granola is mixed with 600 g of Type Y granola to produce a mixture that is 14% nuts by mass. Type X granola has twice as many grams of nuts per gram as Type Y granola. What percentage of nuts is in each type of granola?

3. A pharmacist needs to make 750 mL of a 54% alcohol solution. She has a large bottle containing 1 L of 40% alcohol solution and a large bottle containing 1 L of 70% alcohol solution.

(a) How many mL of each solution should be mixed to make the 54% solution?

(b) What is the maximum volume, rounded down to the nearest mL, of 54% alcohol solution that could be made from the two original 1 L bottles?

4. A jeweller has some 6 carat gold and some 18 carat gold, and wants to produce 120 g of gold that is 60% pure by mass. Pure gold is 24 carat.

(a) What percentage of gold is 18 carat gold?

(b) How many grams of 6 carat gold and of 18 carat gold are needed to produce the mixture?

5. An express train travels 336 km in the same time that a car travels 252 km. The train travels, on average, 28 km/h faster than the car. Find the average speed of the car and the time taken for the car to travel 252 km. *Hint:* let t be the common time, in hours, and write each speed as distance $\div\ t$.

6. A small plane flying into a headwind takes 1.5 h to travel the 450 km from Lethbridge to Edmonton. At the same time, a similar plane leaves Edmonton and reaches Lethbridge in 1.25 h. If the planes have the same cruising speed in windless conditions, determine the speed of the wind.

7. A cyclist leaves home at 7:15 am to bike to the library, 9 km away. She rides at 12 km/h until she gets a flat tire, then walks her bike the rest of the way at 4 km/h. She arrives at the library at 8:10 am. How far did she have to walk her bike?

8. *(Multiple Choice)* A shopkeeper mixes two coffees, one selling at \$9 per kg and the other at \$13 per kg, to make 80 kg of a mixture selling at \$10.60 per kg. The number of kg of the first coffee in the mixture is
A. 20 B. 32 C. 48 D. 60

9. *(Numerical Response)* Aiden left home at 1:00 pm to travel 522 km to visit his uncle. He averaged 95 km/h for the first part of the trip, during which he took a 40-minute rest, and 75 km/h for the second part, during which he took a 20-minute rest. He reached his destination at 8:00 pm. The number of minutes he spent *driving* during the first part of the trip is _____.

Record your answer left to right. | | | | |

Lesson 6 Assignment *Show your work.*

1. *(Multiple Choice)* Solve the system of equations $2x + y = 11,\ \ x - y = 1$ and select the ordered pair (x, y) that satisfies both equations.
A. $(4, 3)$ B. $(3, 4)$ C. $(-4, 3)$ D. $(4, -3)$

2. *(Multiple Choice)* The system $2x - y = 6,\ \ 4y = 8x - 24$ has
A. no solution
B. one solution, $(3, 0)$
C. one solution, $(0, -6)$
D. infinitely many solutions

3. *(Numerical Response)* Solve the system $3x + y = 8,\ \ 2x - 5y = 1$ by substitution or elimination. The y-coordinate of the point of intersection, to the nearest hundredth, is _____.
Record your answer left to right. ☐☐☐☐

4. *(Multiple Choice)* The system $y = 3x - 9,\ \ x = by + c$ has infinitely many solutions if
A. $b = \frac{1}{3}$ and $c = 3$ B. $b = \frac{1}{3}$ and $c = -3$
C. $b = 3$ and $c = 9$ D. $b = 0$ and $c = 0$

5. *(Multiple Choice)* The graphs of $y = ax + b$ and $y = cx + d$ intersect at exactly one point. Which statement must be true?
A. $a = c$ B. $a \neq c$ C. $b = d$ D. $b \neq d$

6. *(Multiple Choice)* When solving a system of equations, one of which is $\frac{x}{3} - \frac{y}{6} = 4$, a substitution which can be made is
A. $x = \frac{1}{2}(y + 4)$ B. $x = \frac{1}{3}(y + 24)$
C. $x = \frac{1}{2}(y + 24)$ D. $x = \frac{1}{6}(y + 24)$

7. *(Multiple Choice)* In solving the system $5a + 2b = 16,\ \ a - 3b = -2$ by elimination, an equation which arises could be
A. $17a = 44$ B. $17a = 48$
C. $13a = 44$ D. $-17a = 44$

8. *(Multiple Choice)* If $x + y = 20$ and $x - y = 6$, then $2x + y$ is equal to
A. 26 B. 33 C. 14 D. 40

9. *(Multiple Choice)* If $3(x-1)-y=10$ and $(x+2)+4y=15$, then x is equal to

A. 5 B. 3 C. $\frac{13}{3}$ D. -5

10. *(Numerical Response)* If $m-4n-15=3m-2n-25=0$, then the value of $m-n$ is _____.

Record your answer left to right. ☐☐☐☐

11. *(Numerical Response)* The line $ax+y=b$ passes through the points $(1,3)$ and $(4,-4)$. The value of ab, to the nearest tenth, is _____.

Record your answer left to right. ☐☐☐☐

12. *(Multiple Choice)* A rowing club rowed 60 km to a campsite. Part of the route crossed a lake into a headwind, where their speed was 5 km/h; the rest was down a river with the current, where their speed was 15 km/h. The whole trip took 6 hours. If x km were rowed into the headwind and y km with the current, which system could be used to determine x and y?

A. $x+y=60,\ \ 5x+15y=6$

B. $x+y=60,\ \ \frac{x}{5}+\frac{y}{15}=6$

C. $x+y=6,\ \ 5x+15y=60$

D. $x+y=6,\ \ \frac{x}{5}+\frac{y}{15}=60$

13. For each system, choose a method (graphing, substitution or elimination) and **justify** your choice. Then solve the system.

(a) $5m-n=13,\ \ 2m+n=8$ (b) $y=2x-7,\ \ 3x+4y=5$ (c) $y=0.35x+1.2,\ \ y=-0.9x+6.2$

14. Two parking garages charge as follows: Garage A charges \$8 flat plus \$1.50 per hour. Garage B charges \$3 flat plus \$2.50 per hour.

(a) Write a linear cost equation for each garage, using y for total cost and x for hours parked.

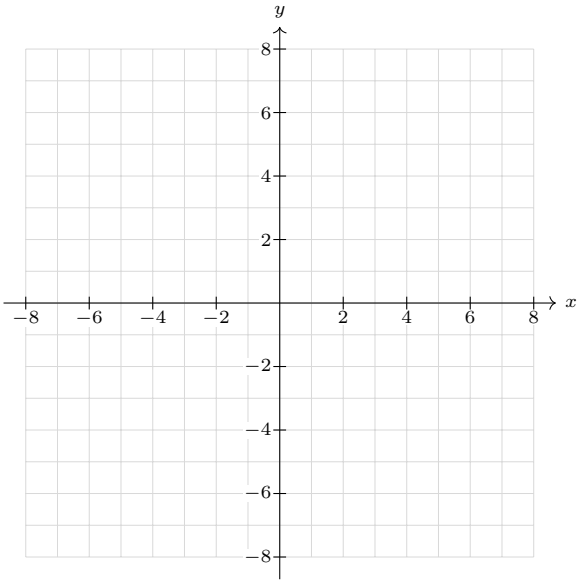
(b) Determine algebraically the number of hours at which both garages cost the same amount.

(c) State which garage is cheaper for a stay of 6 hours, and by how much.

15. Consider the system of equations $x + y = 6$, $2x - y = 3$.

(a) Graph both equations on the grid below without using technology, and state the solution.

Solution: _____



(b) Verify the solution algebraically in both original equations.

(c) Without graphing, state the number of solutions of the system $3x - y = 5$, $y = 3x + 2$, and justify your answer.

16. A chemist mixes a 20% acid solution with a 50% acid solution to obtain 300 mL of a 30% acid solution. Let x be the number of mL of the 20% solution used and y the number of mL of the 50% solution used.

(a) Write a system of two equations modelling this situation — one for the total volume, one for the total amount of acid.

(b) Solve the system.

(c) State how many mL of each stock solution are required, and check your answer against the words of the question.

17. A concession stand sold 240 items in total — hot dogs at \$4.50 each and drinks at \$2.00 each — and took in \$810. Determine how many of each were sold.

18. A rectangular dog run has a perimeter of 84 m. Its length is 6 m more than twice its width. Determine the dimensions of the dog run.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. (a) $y = -\frac{1}{3}x + \frac{1}{3}$, $y = 2x - 9$ (b) $x+3y=1$: $(-5,2), (-2,1), (1,0), (4,-1), (7,-2)$; $2x-y=9$: $(2,-5), (3,-3), (4,-1), (5,1), (6,3)$ (c) $x = 4$, $y = -1$ (d) $4 + 3(-1) = 1$ ✓, $2(4) - (-1) = 9$ ✓. **2.** (a) $x = 3$, $y = 1$ (b) $x = 4$, $y = -8$ (c) $x = -12$, $y = -9$. **3.** (a) $8a + 5s = 2350$ (b) 200 adult, 150 student (c) $200 + 150 = 350$ ✓, $8(200) + 5(150) = 2350$ ✓. **4.** (a) A: $C = 350 + 2.5d$; B: $C = 5d$ (c) (140, 700): a 140 km trip costs \$700 with either company (d) Company B is cheaper for trips shorter than 140 km. **5.** (a) one solution, $(2, 5)$ (b) no solution (c) infinitely many solutions. **6.** (a) no solution (b) one solution (c) infinitely many solutions. **7.** (a) e.g. $5x - y = 8$ (b) e.g. $x - y = 1$ (c) e.g. $10x - 2y = 6$. **8.** A. **9.** 2. **10.** 8.

Lesson 2 Assignment

1. (a) $x = 4$, $y = 3$ (b) $x = -3$, $y = 5$ (c) $x = 2$, $y = -3$ (d) $x = -1$, $y = 4$. **2.** (a) $x = -2$, $y = 3$ (b) $x = 8$, $y = -1$ (c) $p = 1$, $q = -3$ (d) $u = -\frac{5}{2}$, $v = -1$ (e) $x = 1$, $y = 1$ (f) $x = 5$, $y = -2$. **3.** (a) $\left(\frac{3}{2}, 0\right)$ (b) $\left(1, \frac{17}{5}\right)$. **4.** (a) $p + 4q + 10 = 0$, $-2p + 2q + 10 = 0$ (b) $p = 2$, $q = -3$; $2x - 3y + 10 = 0$ (c) $m = \frac{2}{3}$, and $y - 4 = \frac{2}{3}(x - 1)$ rearranges to $2x - 3y + 10 = 0$. **5.** (a) Infinitely many solutions — substitution reduces to $10 = 10$, which is always true; $4x - 2y = 10$ is $y = 2x - 5$ again, so both equations describe the same line. (b) No solution — substitution reduces to $9 = 4$, which is never true; the lines are parallel (both of slope $\frac{1}{2}$) with different y-intercepts. **6.** A notebook costs \$15 (a pen costs \$9). **7.** A. **8.** C. **9.** 1.6. **10.** 7. **11.** (a) $x = 7$, $y = 14$, $z = 1$ (b) $x = -4$, $y = -1$, $z = 1$.

Lesson 3 Assignment

1. (a) $x = 6$, $y = 2$ (b) $x = 7$, $y = -3$ (c) $x = 2$, $y = -4$ (d) $a = 3$, $b = -2$. **2.** (a) $x = 4$, $y = 5$ (b) $m = 3$, $n = 4$ (c) $x = -5$, $y = 3$ (d) $a = -4$, $b = 6$. **3.** (a) $a = 5$, $b = 2$ (b) $x = 3$, $y = 1$ (c) $x = 0.5$, $y = 0.5$. **4.** (a) $x = 3$, $y = 2$ (b) $x = 2$, $y = -3$ (c) $x = \frac{23}{19}$, $y = \frac{32}{19}$. **5.** (a) $x = 2.5$, $y = 0.5$ (b) $x = 4$, $y = 7$ (c) $e = 5$, $f = -3$ (d) $x = 2$, $y = -2$. **6.** $x = 5$, $y = 1$ by both methods; either is reasonable, since the first equation isolates x with no fractions. **7.** $x = 6$, $y = -10$ by both methods; elimination is fastest, since x already has matching coefficients (9). **8.** (a) infinitely many solutions — both equations are the line $x - 3y + 2 = 0$ $\left(y = \frac{1}{3}x + \frac{2}{3}\right)$ (b) no solution — the lines are parallel and elimination leads to a false statement. **9.** (a) $x = 6$, $y = 9$ (b) $m = 9$, $n = 6$ (c) $x = 8$, $y = 3$. **10.** B. **11.** A. **12.** A. **13.** 16.8. **14.** 31.

Lesson 4 Assignment

1. 276 cm^2 (length 23 cm, width 12 cm). **2.** 12 and -5. **3.** \$47.50 (notebook \$3.60, folder \$2.30); shortcut: $10x + 5y = \frac{5}{3}(6x + 3y) = \frac{5}{3}(28.50)$. **4.** length 15 cm, width 10 cm. **5.** \$54 (balcony seat \$27). **6.** 30 nickels, 56 quarters. **7.** \$40 000. **8.** Growth Fund \$25 000, Bond Fund \$15 000. **9.** \$5000 at 8%, \$4000 at 5%. **10.** A. **11.** (a) P (time), Q (output) (b) older machine 8 h and newer machine 10 h — both conditions hold (c) 18 h in total but 590 boxes, so only the time condition holds. **12.** $a = 22.6$, $c = 17.4$: not whole numbers, so the report must contain an error. **13.** 42. **14.** 49. **15.** (Extension) (a) 5.5, 7.7 (b) $a = 0.5$, $b = 0.2$, so $h = 0.5 + 0.2n^2$ (c) $h(5) = 5.5$ and $h(6) = 7.7$, matching part (a).

Lesson 5 Assignment

1. 51 kg of the \$5.50/kg bars and 45 kg of the \$3.50/kg bars. **2.** Type X is 20% nuts; Type Y is 10% nuts. **3.** (a) 400 mL of the 40% solution and 350 mL of the 70% solution (b) 1875 mL. **4.** (a) 75% (b) 36 g of 6 carat gold and 84 g of 18 carat gold. **5.** The car averages 84 km/h and takes 3 hours. **6.** The wind blows at 30 km/h. **7.** She walked 1 km. **8.** C. **9.** 216.

Lesson 6 Assignment

1. A **2.** D **3.** 0.76 **4.** A **5.** B **6.** C **7.** A **8.** B **9.** A **10.** 9 **11.** 12.4 **12.** B **13.** (a) elimination (n-coefficients opposite): $m = 3$, $n = 2$ (b) substitution (y isolated): $(3, -1)$ (c) set the expressions equal (both $y = \ldots$, decimals): (4, 2.6). **14.** (a) $y = 8 + 1.5x$ (A), $y = 3 + 2.5x$ (B) (b) $x = 5$ hours, both \$15.50 (c) Garage A, by \$1 (\$17 versus \$18). **15.** (a) $(3, 3)$ (b) $3 + 3 = 6$ ✓, $2(3) - 3 = 3$ ✓(c) no solution (equal slopes 3, different y-intercepts — parallel and distinct). **16.** (a) $x + y = 300$, $0.20x + 0.50y = 90$ (b) $x = 200$, $y = 100$ (c) 200 mL of the 20% solution and 100 mL of the 50% solution. **17.** 132 hot dogs and 108 drinks. **18.** 30 m by 12 m.